

Shawn Zhao

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EDUCATION

University of Illinois Urbana-Champaign

Master of Computer Science

Dec 2025

GPA: 3.8/4.0

Arizona State University

Bachelor of Science in Computer Science

Dec 2022

GPA: 3.9/4.0

TECHNICAL SKILLS

Languages: Java, Python, Go, JavaScript, TypeScript, C#, C/C++, SQL, HTML/CSS

Frameworks: Spring Boot, Django, Flask, FastAPI, Node.js, Express.js, Next.js, React.js, Selenium, Jenkins

Tools: PostgreSQL, MySQL, MongoDB, SQLite, Redis, Docker, Kubernetes, AWS, GCP

EXPERIENCE

Software Engineer

Jan 2025 – Present

University of Illinois Urbana-Champaign

Champaign, IL

- Containerized satellite network components with Docker and deployed them on Kubernetes to orchestrate and manage 100+ distributed simulation workloads
- Designed and ran large-scale simulations of 100+ dynamic LEO satellites and 20 ground stations in StarryNet to evaluate latency, bandwidth, and routing under realistic orbital motion
- Built a modular BandwidthManager in StarryNet with Python, Docker, and iperf3 (TCP/UDP), supporting 2 adaptive ISL bandwidth allocation modes
- Implemented demand-aware allocation strategies, reducing energy usage by 25% while keeping latency within 10% of baseline across 50+ simulations

Software Engineer Intern

Aug 2024 – Apr 2025

Civilience

Omaha, NE

- Built serverless RESTful APIs with AWS Lambda, Python (Flask), and Node.js (Express), improving average response times by 22%
- Migrated legacy APIs to microservices architecture, decreasing request latency from 150ms to 105ms and saving \$1,200 annually in server costs
- Secured user authentication with AWS Cognito and managed 500K+ records across DynamoDB and RDS, improving data access latency by 35ms
- Automated 30+ IAM role assignments for AWS EC2 and Amplify using Terraform scripts, saving 30 hours of manual setup time per month

Software Engineer

Feb 2023 – Jul 2024

Rnd4impact

San Jose, CA

- Built notification and audit logging microservices in Python (Django) with PostgreSQL, delivering real-time alerts with under 1s latency for 7K+ daily events
- Developed file upload and document management microservices in Java (Spring Boot) with AWS RDS and S3, supporting reliable processing of 1K+ files per week
- Built a responsive frontend using React (Next.js) and TypeScript, reducing load time by 800ms
- Automated regression testing with Selenium and Jenkins, saving 5+ hours of QA time per release and preventing 20+ bugs from reaching production

PROJECTS

AI Log Analyzer | Python, FastAPI, PostgreSQL, Redis, Docker, OpenAI API

Jan 2026 – Apr 2026

- Built an AI backend service that ingests 10K+ simulated logs/day and generates structured incident reports
- Created REST APIs for log ingestion, incident search, service filtering, and historical analysis
- Implemented Redis async processing, error grouping, caching, and validation, cutting repeated LLM calls by 30%

RainStorm | Java, Distributed Systems, Multithreading, Socket Programming

Aug 2024 – Dec 2024

- Built a membership-list protocol with gossip heartbeats, leader election, and node tracking for P2P coordination
- Developed a multithreaded P2P file system with chunk replication and dynamic discovery for reliable data transfer
- Implemented a real-time Java stream processing system with optimized pipelines (+15% throughput), 2s failure detection/recovery, and dynamic task rescheduling